Thijmen **Nijdam**

Amsterdam, Netherlands · *thijmen.nijdam@gmail.com* · *+31629375397*

Homepage | GitHub | LinkedIn | Google Scholar



Professional Summary

Robot Learning Research Engineer at the University of Amsterdam with full ownership of a real-to-sim-to-real pipeline for robot manipulation, in collaboration with Toyota Motor Europe and the Toyota Research Institute. Experienced in imitation learning and VLAs on real hardware and in training on multi-GPU HPC clusters, using Python, PyTorch, ROS 2, and LeRobot. MSc thesis on world models for robot planning with model predictive control. Co-authored six publications, including papers at ICML and SIGIR.

Experience

Robot Learning Research Engineer, *University of Amsterdam* **May 2026 - Present**

- Training and deploying VLA and imitation learning policies on real hardware, with successful rollouts from a single demonstration
- Working daily with a Franka Research 3 arm, owning the full robotics stack from robot control and data processing to deployment
- Personally collecting teleoperated demonstrations and running real-world evaluations, chasing down policy failure modes
- Closing the sim-to-real gap by aligning a digital twin of the real scene, so that policies fine-tuned with RL transfer to the real robot

Research Intern, *University of Amsterdam* **Jan. 2024 - Feb. 2026**

- MSc thesis: designed and trained a local world model that opens unseen drawers and doors via MPC (RSS Workshop 2026)
- Built segmentation and tracking pipelines to benchmark physical reasoning of video generation models (ICML 2026)
- Benchmarked 40+ information retrieval models including LLM re-rankers and fine-tuned models on negation (SIGIR 2025)
- Developed a conformal prediction method for uncertainty quantification in time series forecasting (COPA 2024)

AI Developer, *LeerLevels* **Apr. 2022 - Mar. 2026**

- Built a RAG-based assistant helping teachers assemble lessons and edit educational content
- Deployed an LLM-powered grading feature with monitoring via Langfuse, providing tailored feedback to hundreds of students

AI Engineer, *RoboCup Student Robotics Team* **Feb. 2023 - Jan. 2024**

- Implemented a real-time ball and robot detection pipeline (Single Shot Detector) on NAO humanoid robots
- Built a whistle detection model for referee signal recognition

Education

MSc Artificial Intelligence **Sep. 2023 - Feb. 2026**

University of Amsterdam

- Grade: 8.3
- Relevant courses: Computer Vision, Reinforcement Learning, Foundation Models, Machine Learning I & II, Deep Learning I & II

BSc Artificial Intelligence **Sep. 2020 - Jun. 2023**

University of Amsterdam

- Grade: 8.1, graduated with honours
- Minor in computer science, exchange at the University of Hong Kong

Selected Publications

Local World Models for Articulated Object Manipulation **RSS Workshop**

Robot World Models Workshop @ Robotics: Science and Systems | *First author* 2026

- Paper | Website

Evaluating Newtonian Mechanics in Video Generative Models with Real Physical Systems **ICML**

International Conference on Machine Learning | *Co-second author* 2026

- Paper | Website | Code

Reproducing NevIR: Negation in Neural Information Retrieval **SIGIR**

ACM SIGIR Conference on Research and Development in Information Retrieval | *Co-first author* 2025

- Paper | Code | Poster

Conformal Time Series Decomposition with Component-wise Exchangeability **COPA**

13th Symposium on Conformal and Probabilistic Prediction with Applications | *Co-first author* 2024

- Paper | Code | Slides

Extracurricular Activity

- World Models Reading Group: member of a reading group on world models for robotics, occasionally presenting papers Present
- AI Safety Hackathon: improving LLM interpretability, achieving 2nd place 2024
- European Robotics Forum Hackathon: implemented autonomous driving and object detection on a wheeled robot 2022